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Online simulation versus traditional classroom learnings in clinical pharmacy education: effect on students' knowledge, satisfaction and self-confidence

Aysu Selcuk^{1*}, Nur Ozturk², Nurbanu Onal³, Asuman Bozkir⁴ and Nilay Aksoy⁵

Abstract

Background Over the course of the past few years, the area of medical education has experienced a substantial movement towards the establishment of online learning platforms and resources. This study aimed to to evaluate the efficacy of an online simulation learning intervention, MyDispense®, compared to traditional classroom learning in terms of enhancing knowledge, satisfaction, and self-confidence among participants.

Methods A multicentre randomized controlled study was conducted among pharmacy students who were assigned either intervention MyDispense® or control traditional classroom learning groups. They were eligible if they previously had experience with online simulation learning. A previously validated questioner were used to measure the outcome of knowledge, satisfaction and self-confidence.

Results Both the intervention and control groups revealed significant improvement in knowledge, the P value for pre-post knowledge scores for each group was < 0,001. Despite these internal improvements, this study's findings showed no statistically significant differences ($p > 0.05$) between the intervention and control groups on knowledge gain, satisfaction, or self-confidence. This represents comparable outcomes irrespective of the group's exposure to intervention.

Conclusion The study evaluated the efficacy of online simulation learning intervention MyDispense® in comparison to traditional classroom learning. While both strategies effectively improved knowledge, satisfaction, and self-confidence, the findings demonstrated that the online simulation yielded equivalent learning benefits. MyDispense® could be an alternative to traditional education in situations where face to face learning is not feasible, with comparable learning outcomes.

Clinical trial number not applicable.

Keywords Clinical pharmacy education, Self-confidence, MyDispense, Online simulation learning, Traditional classroom learning

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Background

Over the course of the past few years, the area of education has experienced a substantial movement towards the establishment of online learning platforms and resources. The adaptability and convenience of online pharmacy education have contributed to its growing popularity in recent years [1]. However, traditional face to face pharmacy education comes with its own set of benefits. In the context of pharmacy education, face to face refers to the traditional classroom-based learning environment, where students physically attend laboratories, lectures and discussions in the university campuses. This type of education enables direct interaction between students and instructors, allowing students to provide real-time feedback and seek clarification on various concepts. Additionally, traditional classroom-based learning encourages students to work together via activities such as group projects and case studies, which helps students improve their ability to solve problems and communicate effectively [2]. Face to face instruction in pharmacy programmes often includes hands-on training in compounding, dispensing, and patient counselling. This helps students get ready for the practice that they will encounter in the real world [3]. In contrast, online simulation based pharmacy education utilises digital platforms as a means to disseminate course materials and enable virtual interactions. Online programmes provide students with the advantage of flexibility in terms of the timing and location of lectures and assignments, therefore accommodating a wide range of learners with various schedules [4]. The availability of digital materials facilitates self-directed learning and caters to persons who have a preference for self-directed study approaches. While online education provides advantages in terms of flexibility and convenience, it may lack the interpersonal engagement necessary for fostering a deeper understanding of complex pharmacy cases [5]. In the context of advanced clinical pharmacy, simulation has been demonstrated to play a significant role in enhancing pharmacotherapeutic knowledge and developing patient-oriented care skills among pharmacists. Additionally, it has also been demonstrated to boost students' confidence in handling exceptional situations and unforeseen emergencies [6].

The integration of MyDispense® programme, which is an online virtual simulation platform, into traditional learning curriculum has the potential to greatly advance pharmaceutical education by bridging the gap between virtual and traditional learning settings [7]. As an example of online pharmacy education, *MyDispense®* provides valuable insights into the potential of virtual platforms to train aspiring pharmacists [8]. It offers pharmacy students an immersive and comprehensive learning experience through a variety of features such as virtual simulations and interactive modules [9]. The platform

allows students to practise skills in a virtual environment, preparing them for real-world situations in a controlled setting. Although MyDispense® and other online pharmacy education platforms provide several benefits, it is critical to address all the challenges and concerns. The aforementioned factors include the need for stable internet access, and promoting effective student engagement within a virtual educational context [10]. Online simulation tools like MyDispense® are not intended to supplant classroom learning but rather to supplement it by providing students with opportunities to practice and refine their skills in a controlled, flexible, and resource-efficient environment, thus they can also perform certain functions better than face to face learning, such as providing scalable, personalized practice or simulating complex scenarios which may be hard to replicate in real life settings.

Ultimately, the effectiveness of any educational modality is determined by the student's particular features and chosen techniques of learning. Education must recognise the benefits of combining face to face and online education to provide future pharmacist with a full learning experience [11]. Moreover, there is no clear consensus on whether face to face education can be fully replaced by online learning or how a hybrid approach combining both methods should be implemented. It is essential to evaluate the advantages and disadvantages of each in terms of gain in knowledge, satisfaction or self-confidence to effectively integrate their strengths and enhance education. The purpose of this study is to acquire insight into the effectiveness of online simulation based learning compared to traditional classroom based learning in clinical pharmacy education by evaluating students' knowledge, satisfaction and self-confidence. The results of this study will highlight the potential benefits and drawbacks of integrating online simulation into clinical pharmacy curricula.

Methods

Study design, settings and recruitment

This was a multicentre and randomized controlled study conducted in a governmental school (Ankara University) and private school (Altinbas University), Faculties of Pharmacy in Türkiye. Pharmacy students who had previously received online simulation training were invited to participate voluntarily in the study. Both universities provides training in 4th and 5th years of the curricula. Therefore only 5th year students were deemed eligible for the study. An informed consent form was obtained from the students at the beginning of the study.

Students who agreed to participate in the study were randomly assigned to either the traditional classroom control group or the online simulation *MyDispense®* intervention group. A simple randomization process was

used with support from an algorithm created by Research Randomizer[®]. To avoid bias, each student was assigned a unique number, and based on these numbers, they were allocated to either the control or intervention group. The control group comprised students who received traditional classroom-based learning, where a lecturer presented topics along with relevant videos and facilitated case discussions. The intervention group comprised students who received online simulation learning through the MyDispense[®] programme. These involved students solving cases by utilizing materials provided by the lecturer and utilizing the platform's features. The research team created identical case materials for both the control and intervention groups. The case involved a young woman who was prescribed pills for oral contraceptives.

To prevent learning loss during the intervention, pre- and post-assessments were conducted to track progress. During the intervention, monitoring and feedback mechanisms were implemented to identify and address any learning gaps. To ensure a clear and unbiased comparison of the two methods, the face to face group did not have access to MyDispense[®], and the MyDispense[®] group did not receive face to face learning. Additionally, there was no communication between the two groups to avoid cross-contamination of learning experiences.

The sample size was calculated using the G*Power[®] 3.1 software. The effect size ($d: 0.50$), margin of error ($\alpha = 0.05$), and power ($1 - \beta = 0.80$) were used to calculate the number of samples for each group, which was determined to be 64.

Outcome measures

The main outcome assessed in the study were knowledge, satisfaction, and self-confidence. Expert pharmacotherapy professors developed the knowledge assessment questions. The knowledge survey comprised a total of 10 questions, encompassing both multiple-choice and short-answer formats, all related to the case. A pilot study was conducted to measure content validity of the knowledge survey among the pharmacy students. The survey questions were revised based on the feedback from the

students. Students who participated in the pilot study were not included in the study. The knowledge survey was implemented before and after the traditional classroom learning and online simulation learning.

Student Satisfaction and Self-Confidence in Learning Survey was used, and it contains 13 questions and is a 5-point Likert type survey [12]. Cronbach's alpha value of this survey was reported as 0.94 for the satisfaction subscale and 0.87 for the self-confidence subscale [13, 14]. The Turkish validity and reliability of the survey was conducted and the Cronbach's alpha value was found to be 0.95 [15]. The survey was implemented once after the education. Students who did not complete all the surveys were excluded from the study.

Ethics

This study was approved by the Research Ethics Committee of Ankara University (Approval no: 09–68, date: 29.05.2023).

Statistical analysis

Statistical analysis was performed using SPSS[®] (Statistical Package for Social Sciences) Version 21.0. Categorical variables were described as percentages, and continuous variables were expressed mean (standard deviation) (SD) or median (interquartile range (IQR)). Normal distribution was tested among continuous variables. Chi-square test, Wilcoxon (paired) signed rank test and Mann Whitney U test were used to evaluate significant differences between study groups. The results were evaluated at the 95% confidence interval and the significance level at $p < 0.05$.

Results

A total of 141 students were eligible to participate in the study ($n = 71$ intervention, $n = 70$ control). However, 7 students were excluded as they did not complete the surveys ($n = 3$ intervention, $n = 4$ control) (Fig. 1: **Recruitment and randomisation process of the students**).

Of the participants, 76% were female, and all were in their 5th year of study. Approximately 41% of the students

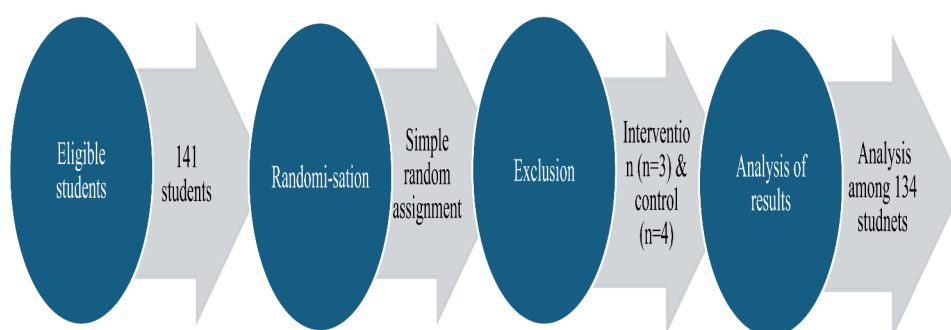


Fig. 1 Recruitment and randomisation process of the students

Table 1 Baseline characteristics of the students

Characteristics	Inter- vention (n = 68)	Control (n = 66)	All stu- dents (n = 134)
Gender, female, n (%)	52 (77)	50 (76)	102 (76)
Age, median (IQR)	25 (24–25)	24 (23–25)	24 (24–25)
University, n (%)			
Altinbas University	41 (60)	38 (58)	79 (59)
Ankara University	27 (40)	28 (42)	55 (41)
Year of pharmacy education, 5th year, n (%)	68 (100)	66 (100)	134 (100)
Received any education about the case before the study, n (%)	55 (81)	48 (73)	103 (77)

Table 2 Comparison of knowledge, satisfaction and self-confidence between the intervention and control groups

Characteristics	Intervention (n = 68)	Control (n = 66)	P value
Knowledge, median (IQR)			
Pre-learning	6 (4–7)	5 (4–7)	0.016
Post-learning	7 (6–9)	7 (6–8)	0.402
P value	< 0.001*	< 0.001*	
Satisfaction	20 (19–23)	21 (20–24)	
Self-confidence	32 (31–35)	32.5 (20–34)	0.243

* p value < 0.05 considered significant

were enrolled at Ankara University, while the remaining 59% were attending Altinbas University. The median (IQR) of the students was 24 (24–25). Approximately 77% were educated about oral contraceptives during a particular lecture in their pharmacy curriculum (Table 1).

The median (IQR) pre-knowledge scores were 5 (4–6) at Ankara University and 6 (4–8) at Altinbas University, with a p-value of < 0.001. After the intervention, the median (IQR) post-knowledge scores increased to 7 (6–8) at Ankara University and 7 (6–9) at Altinbas University, also with a p-value of < 0.001 (Table 2).

Both the intervention and control groups revealed significant improvement in knowledge, the P value for pre- post knowledge scores for each group < 0.001. The satisfaction score [Median (IQR) = 20 (19–23) versus 21 (20–24)], and self-confidence score [Median (IQR) = 32 (31–35) versus 32.5 (20–34)] did not show any significant differences between the students in the intervention group (MyDispense®) and those in the control group (traditional classroom). The p-values for these comparisons were 0.456 and 0.243, respectively (Table 2).

Discussion

The ongoing global challenges due to various viral outbreaks, from COVID-19 to the newest, just-emerging monkeypox virus, have changed the functioning of societies in very different ways, especially education. These

health crises put up an atmosphere of uncertainty and caution that has compelled educational systems worldwide to rethink traditional methods. All the world witnessed increased online education, for instance, where schools and universities resorted to closing doors for physical learning in a bid to be able to curb the spread of the coronavirus [16]. What was initially considered a temporary solution has now evolved into a more permanent transformation as the benefits of digital learning, such as accessibility and flexibility have become increasingly apparent. MyDispense®, an online simulation application, has been awarded the 2022 Australian Financial Review Higher Education Award for Teaching and Learning Excellence. Monash University successfully distributed this programme globally, with students from over 40 countries and 220 institutions utilizing it (<https://www.monash.edu/pharm/teaching-learning/mydispense>). The first Turkish version of the programme was jointly designed by Altinbas University and Monash University, with the collaboration eventually expanding to include Ankara University [9]. In this study, we aimed to evaluate the efficacy of an online simulation learning intervention, MyDispense®, compared to traditional classroom learning in terms of enhancing knowledge, satisfaction, and self-confidence among participants. When comparing simulation based MyDispense® to traditional classroom learning in pharmacy education, it is important to consider several elements, including effectiveness, practicality, cost, and student participation. Our research set out to compare two teaching strategies used by students who were able to effectively resolve oral contraceptive drug issues, using a questionnaire, we measured students' knowledge, satisfaction, and self-confidence. Our study's main findings can be separated into three major outcomes, as shown below.

Demographic differences

The majority of students in the study were of similar age and were in their fifth year of pharmacy education. There were no notable disparities in pre knowledge scores observed among students from private and governmental universities. This could be attributed to the strong collaboration between the two universities, where instructors collaborated on the educational foundation and shared experiences and cases related to the MyDispense® programme. This collaboration resulted in the establishment of standardized education in these two facilities. This finding is corroborated by two prior studies that have indicated that collaboration among educators has the potential to boost student learning through the facilitation of idea exchange and collaborative efforts in the context of MyDispense® cases [3, 17].

Knowledge acquisition and the intervention effect

The findings of this study on knowledge acquisition are somewhat equivocal. MyDispense® and traditional classroom groups evidenced statistically significant increases in knowledge post-learning within the two groups. This indicates that the standardized education provided at both universities enabled students to achieve similar knowledge levels, despite initial differences in pre-knowledge scores. Both teaching strategies were found to be equally effective in improving students' understanding of oral contraceptives, indicating that *MyDispense*® could serve as a viable alternative for teaching pharmacotherapy cases. However, it is important to acknowledge the strengths and limitations of each approach.

Traditional classroom learning provides several advantages, such as direct interpersonal engagement, spontaneous learning opportunities, and the development of interpersonal skills through face to face instruction. Within the realm of pharmacy education, traditional education provides students the opportunity to gain practical experience in authentic pharmacy environments, enabling them to engage with patients, healthcare practitioners, and the tangible surroundings of a pharmacy. This form of education additionally facilitates the cultivation of communication aptitude, problem-solving proficiencies, and decision-making proficiencies through direct interpersonal engagement and hands-on encounters. Moreover, traditional educational approaches might offer students impromptu learning chances that may not be completely duplicated in virtual or simulated settings [18, 19]. However, virtual online simulation programmes such as MyDispense® offer advantages in terms of accessibility and convenience. As an online or digital platform, it could provide flexibility for students to learn at their own place and on their own schedule [20]. The advocate of traditional classroom education exhibits a reluctance to incorporate modern technology and devices within educational institutions, perceiving them as potential sources of diversion rather than valuable tools for enhancing learning [18, 19]. There are conflicting results in the literature about the role of simulation learning in pharmacy education. A study conducted by Lim et al., showed that there was no significant association between the number of online practice attempts made by students and their exam grades [21]. In a scoping review conducted by Khera et al. (2023), forty-three papers were analysed to show how MyDispense® is supporting pharmacy education, skill development, and application of knowledge and student outcomes [22]. The study revealed that MyDispense® had a crucial role in improving students' dispensing skills, communication skills, and decision-making/problem-solving skills within the context of skill development [22].

Additionally, another study highlighted that MyDispense® enables the practical implementation of theoretical knowledge, particularly in areas such as pharmacy laws [23], principles of medication theory, and comprehension of prescription drugs and non-prescription items. MyDispense® additionally facilitates educators by offering a secure setting for students to cultivate diverse pharmacy competencies, obtain timely evaluations, and participate in introspective learning [3, 24, 25, 26]. The programme has proven to be quite beneficial in facilitating instruction and fostering the growth of student skills, notably in the face of the COVID-19 pandemic when conventional teaching approaches were disrupted. A prior study conducted in Türkiye demonstrated that MyDispense® can provide significant assistance to pharmacy education by offering a secure setting for students to engage in practice and cultivate crucial skills necessary for pursuing a career as pharmacists [9]. Furthermore, it can assist educators in providing hands-on instruction and enhancing student learning outcomes. Moreover, the software's modular architecture and capacity for customization suggest its adaptability to the unique requirements of pharmacy education in Türkiye [9].

Satisfaction and self-confidence

In our study, no statistically significant differences were found in terms of students' satisfaction and confidence between simulation-based education MyDispense® and traditional classroom learning. The use of MyDispense® did not reduce satisfaction or self-confidence among participants. High satisfaction levels are associated with continued engagement and motivation to learn, while self-confidence is key in how students apply knowledge. The fact that these aspects remained intact suggests that the intervention could be a plausible substitute for conventional means, divergent in approach, and without loss of subjective quality in the learning experience.

According to numerous studies, the use of MyDispense® in pharmacy education has had a significant impact on student satisfaction and confidence. A scoping review revealed that multiple articles emphasized the positive impact of students' confidence, statistification and proficiency in different facets of the pharmacy profession, including dispensing, clinical practice, and outpatient pharmacy practice [22]. Moreover, the secure setting provided by MyDispense® allowed students to engage in practice and enhance their understanding without the fear of causing harm to the patients or facing real-world consequences. This feature was seen as highly beneficial for students and played a significant role in enhancing their overall satisfaction with the educational experience [9, 23, 26, 27].

A study comparing the feasibility and acceptability of a simulation based and face to face to face learnings among

junior doctors found that online simulation based learning feasible and acceptable, but not rated as highly as face to face teaching [28]. Similarly, a systematic review about nursing education states that online simulation is widely accepted by students and seems to offer learning benefits comparable to other simulation methods while enhancing face to face instruction [29]. In another study among pharmacy students, those participated in online simulation based learning had generally greater improvements in their clinical knowledge score. Third-year students also found the simulation fun and engaging. Similar to our study, online simulation based learning provided an educational experience as effective as the face to face education [30].

Strength and limitation of the study

One of the strengths of our study was the underlining of the measurement of knowledge acquisition and affective dimensions of learning, such as satisfaction and self-confidence. Such prominent factors in the process of learning need to be accounted for when designing and evaluating educational interventions. Moreover, rigorous empirical evaluation is required to firmly establish how interventions influence learning outcome measures; our study had a control group used to establish that the observed effects were the result of the intervention. This study designated the endpoint for assessing self-confidence between the two groups to evaluate the cumulative impact of learning on students' self-confidence. Nevertheless, this cannot overlook the reality that self-confidence may fluctuate based on the degree of knowledge acquisition; so, future research should assess self-confidence before and after learning to observe any changes. To provide a comprehensive comparison between the two educational strategies, it would also be beneficial to analyze the perspectives of instructors. While our study focused on students' satisfaction with the *MyDispense*[®] programme, it did not include instructors' satisfaction or the challenges they faced. Preparing cases for the *MyDispense*[®] programme may be time-consuming, complex, or challenging for instructors. We plan to address this limitation in a subsequent study to gain a more holistic understanding of the topic. Another limitation of our study is the sample size, which, while sufficient for the two universities involved, may not allow for generalization of the findings to other universities in Türkiye. Due to the widespread usage of the *MyDispense*[®] simulation programme by Altinbas University and Ankara University, our study exclusively included students from these two institutions.

Conclusion

The study evaluated the efficacy of online simulation learning intervention *MyDispense*[®] in comparison to traditional classroom learning. While both strategies effectively improved knowledge, satisfaction, and self-confidence, the findings demonstrated that the online simulation yielded equivalent learning benefits. *MyDispense*[®] could serve as a variable alternative to traditional education in situations where face to face learning is not feasible, with comparable learning outcomes. Future research should aim to explore the factors influencing the effectiveness of the intervention, provide a more comprehensive understanding of its impact on different learners, and incorporate instructors' perspectives and challenges to further enhance its implementation and utility.

Abbreviations

SPSS	Statistical Package for Social Sciences
SD	Standard Deviation
IQR	Interquartile Range

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Author contributions

All authors contributed to the study's conception and design. The first draft of the manuscript was written by [A.S], [N.A], [N.O]. All authors read and approved the final manuscript.

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Data availability

The data are available upon a request from the corresponding author.

Declarations

Ethics approval and consent to participate

This study was approved by the Research Ethics Committee of Ankara University (Approval no: 09–68, date: 29.05.2023) and adhered to the principles of the Helsinki Declaration. An informed consent form was obtained from the students at the beginning of the study.

Competing interests

The authors declare no competing interests.

Consent for publication

Not applicable.

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